

Branton DeMoss

Contact	branton.demoss@maths.ox.ac.uk www.brantondemoss.com +44 (0)7926 576225	Mathematical Institute Woodstock Road, Oxford OX2 6GG, UK
Education	<i>DPhil Machine Learning</i> University of Oxford	2025
	<i>BA Mathematics and Physics</i> University of Colorado Boulder	2018
	<i>Visitor Mathematical and Theoretical Physics</i> University of Oxford	2016-17
Experience	Mathematical Institute, University of Oxford <i>Postdoctoral Research Associate</i> • Research on the mathematical and computational foundations of AI. • Erlangen AI Hub affiliate.	2025-
	Oxford Robotics Institute <i>Graduate Student Researcher</i> • Research in complexity, generalization, reinforcement learning, world models.	2021-25
	The Collaboratory <i>Co-founder; Chief Science Officer</i> • Deep learning on language and graphs for knowledge curation. • Led product strategy, design, and ML R&D.	2020-23
	Comma.ai <i>ML Research Intern</i> • Reinforcement learning for self-driving cars.	2020
	Front Range Geosciences <i>Machine Learning Engineer</i> • Sole dev on ML system for seismic data, sold to multinational co.	2017-20
	Center for Theory of Quantum Matter <i>Research Assistant</i> • Research on quantum many-body localization.	2017
	Mathematics Department, CU Boulder <i>Research Assistant</i> • Knot theory and topological quantum field theory.	2016
	High Energy Particle Physics Group, Physics Department, CU Boulder <i>Research Assistant</i> • High performance Monte Carlo simulations (C++) for DUNE experiment.	2014-15
Publications	<i>Geometry, Generalization, and Effective Dimensions</i> Work in progress.	2026

	<i>Escaping Parameter Space: Tight Generalization Bounds via Representation Quality</i> 2026 Under submission to NeurIPS	
	<i>Complexity Dynamics of Learning Systems</i> Doctoral Thesis	2025
	<i>The Complexity Dynamics of Grokking</i> Physica D: Nonlinear Phenomena	2025
	<i>LUMOS: Language-Conditioned Imitation Learning with World Models</i> ICRA 2025	2024
	<i>DITTO: Offline Imitation Learning with World Models</i> arXiv:2302.03086	2023
	<i>Combining physics and deep learning to automatically pick first breaks in the Permian Basin</i> First International Meeting for Applied Geoscience & Energy	2021
	<i>Ein Liebesbrief an KataGo</i> Deutsche Go Zeitung, Ausgabe 4/2020	2020
	<i>Love Letter to KataGo, or: Go AI Past, Present, and Future</i> American Go E-Journal	2020
	<i>DeepTrace: A breakthrough application of deep learning to automate first break picking</i> SEG 2019 Lenovo Thought Leadership Series	2019
	<i>Topology and Knot Theory</i> Course notes for CU Boulder special topics course: “Topology, Knot Theory, and their applications in Physics and Chemistry”	2016
	<i>Secondary Particle Showers from Hadron Absorber Interactions</i> Deep Underground Neutrino Experiment (DUNE) Collaboration Documents	2016
Service and Teaching	<i>Third Symposium on Algorithmic Information Theory and Machine Learning</i> Co-organizer, July 2026	2026
	<i>Theoretical Physics for Machine Learning</i> Oxford, HT 26	2026
	<i>Physics of Information and Complexity</i> Received highest possible marks for teaching performance. Oxford, HT 24	2024
	<i>Philosophy of Emergence</i> Received highest possible marks for teaching performance. Oxford, HT 24	2024

	<i>Topics in Reinforcement Learning</i>	2023
	Received highest possible marks for teaching performance. Oxford, MT 23	
	<i>Rocket League Behaviour Cloning from Unlabelled Data</i>	2023
	Supervised Master's Thesis, Oxford Student obtained highest marks, and secured funded DPhil position in Oxford.	
Talks	<i>Erlangen AI Hub</i>	2026
	Talk on complexity, geometry, and generalization in deep neural networks.	
	<i>Erlangen AI Hub</i>	2025
	Talk on information phase transitions in neural networks, and related gauge-theoretic issues.	
	<i>2nd Symposium on Algorithmic Information Theory and Machine Learning</i>	2025
	Talk on my discovery of complexity phase transitions in learning systems. Link .	
	<i>ICRA 2025, Robot Foundation Models Session</i>	2025
	Talk on our work LUMOS, addressing reinforcement learning in world models.	
	<i>Harvard/Tufts, Levin Group</i>	2025
	Invited talk on complexity dynamics to Michael Levin's computational biology group. Link .	
	<i>Oxford, Department of Physics</i>	2024
	Invited talk on complexity dynamics to Ard Louis's research group.	
	<i>Oxford, Department of Statistics</i>	2024
	Invited talk on complexity and generalization to the RainML group.	
Awards	<i>AWS Lighthouse Scholarship (fully funded PhD)</i>	Oxford, 2021
	<i>Stribic-Martin Scholarship (\$10k for top mathematics undergrad)</i>	Boulder, 2017
	<i>UROP Fellowship (\$3k summer research grant)</i>	Boulder, 2017
	<i>Dawkins Fund Award (Exeter College)</i>	Oxford, 2016
	<i>Gilman Scholarship (\$5k US Dept of State)</i>	Oxford, 2016
	<i>Esteemed Scholar Award (\$7k/yr for top enrolling undergrads)</i>	Boulder, 2014
References	<i>Prof. Nick Hawes</i> Professor of AI and Robotics, Oxford Director, Oxford Robotics Institute nickh@robots.ox.ac.uk	
	<i>Prof. Ingmar Posner</i> Professor of Applied AI, Oxford Deputy Director, Oxford Robotics Institute Associate Head of Engineering Dept. (Research) ingmar@robots.ox.ac.uk	
	<i>Prof. Jakob Foerster</i> Associate Professor, Machine Learning, Oxford jakob@robots.ox.ac.uk	